# CH 6 - Eigenvectors and Diagonalization

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## Matrix of a Linear Mapping, Similar Matrices

Recall the derivation for the standard matrix $[L]$ of a linear mapping $L : \mathbb{R}^n \to \mathbb{R}^n$ for an arbitrary $\vec{x} \in \mathbb{R}^n$, we have

$$L(\vec{x}) = L(x_1\vec{e}_1 + ... + (x_n\vec{e}_n)) = x_1L(\vec{e}_1) + ... + x_nL(\vec{e}_n) = ... = [L]\vec{x}$$

Now we wish to generate an arbitrary basis $B = \{\vec{v}_1, ..., \vec{v}_n\}$ of $\mathbb{R}^n$.

Now the goal is to find a matrix $[L]_B$ s.t.

$$[L(\vec{x})]_B = [L]_B[\vec{x}]_B$$

$$\begin{aligned}
[L(\vec{x})]_B &= [L(b_1\vec{v}_1 + ... + b_n\vec{v}_n)]_B, \quad b_1, ..., b_n \in \mathbb{R} \\
&= [b_1L(\vec{v}_1) + ... + b_nL(\vec{v}_n)]_B \\
&= b_1[L(\vec{v}_1)]_B + ... + b_n[L(\vec{v}_n)]_B \\
&= \begin{bmatrix}[L(\vec{v}_1)]_B & ... & [L(\vec{v}_n)]_B\end{bmatrix} \begin{bmatrix} b_1 \\ ... \\ b_n \end{bmatrix} \\
&= [L]_B[\vec{x}]_B
\end{aligned}$$

> 🐳 **Info — $B$-Matrix**
>
> Let $B = \{\vec{v}_1, ..., \vec{v}_n\}$ be a basis for $\mathbb{R}^n$ and let $L : \mathbb{R}^n \to \mathbb{R}^n$ be a linear operator. The $B$-matrix of $L$ is defined as
>
> $$[L]_B = \begin{bmatrix}[L(\vec{v}_1)]_B & ... & [L(\vec{v}_n)]_B\end{bmatrix}$$
>
> It satisfies $[L(\vec{x})]_B = [L]_B[\vec{x}]_B$

Examples:

Let $\vec{a} = \begin{bmatrix}5 \\ 3\end{bmatrix}, n = \begin{bmatrix}-3 \\ 5\end{bmatrix}$

Let $B = \{\vec{a}, \vec{n}\}$ for $\mathbb{R}^n$

Let $L : \mathbb{R}^2 \to \mathbb{R}^2$ be the mapping given by $L(\vec{x}) = \text{proj}_{\vec{a}}(\vec{x})$

1. $[L]$

   $L(\vec{e}_1) = \text{proj}_{\vec{a}}(\vec{e}_1) = \frac{\vec{e}_1 \cdot \vec{a}}{\|\vec{a}\|^2}\vec{a} = \frac{5}{34}\begin{bmatrix}5 \\ 3\end{bmatrix} = \begin{bmatrix}\frac{25}{34} \\ \frac{15}{34}\end{bmatrix}$

   $L(\vec{e}_2) = \text{proj}_{\vec{a}}(\vec{e}_2) = \frac{\vec{e}_1 \cdot \vec{a}}{\|\vec{a}\|^2}\vec{a} = \frac{3}{34}\begin{bmatrix}5 \\ 3\end{bmatrix} = \begin{bmatrix}\frac{15}{34} \\ \frac{9}{34}\end{bmatrix}$

   $[L] = \begin{bmatrix}\frac{25}{34} & \frac{15}{34} \\ \frac{15}{35} & \frac{9}{34}\end{bmatrix}$

2. $[L]_B$

$L(\vec{a}) = \vec{a} = \vec{a} + 0\vec{n}$

$= [L(\vec{a})]_B = \begin{bmatrix}1\\0\end{bmatrix}$

$L(\vec{n}) = \vec{0} = 0\vec{a} + 0\vec{n}$

$= [L(\vec{a})]_B = \begin{bmatrix}0\\0\end{bmatrix}$

$[L]_B = \begin{bmatrix}1 & 0\\0 & 0\end{bmatrix}$

> **Info — Diagonal Matrix**
>
> A $n \times n$ matrix $D$ is said to be a diagonal matrix if $d_{ij} = 0, \forall i \neq j$. Denoted
>
> $$\text{diag}(d_{11}, d_{22}, ..., d_{nm})$$

**Similar Matrix**

Let $A = [L] = [L(\vec{e}_1) \ ... \ L(\vec{e}_n)]$

Let $B = [L]_B$

Let $P = {}_S P_B = [\vec{v}_1 \ ... \ \vec{v_n}]$ where $S = \{\vec{e}_1 \ ... \ \vec{e}_n\}$

Note that $[x]_S = {}_S P_B [\vec{x}]_B \implies \vec{x} = P[\vec{x}]_B \implies [\vec{v}]_B = P^{-1}\vec{x}$

$B = [L]_B = \left[[L(\vec{v}_1)]_B \ ... \ L(\vec{v}_n)\right]_B$

$= \left[[[L]\vec{v}_1]_B \ ... \ [[L]\vec{v}_n]_B\right]$

$= \left[[A\vec{v}_1]_B \ ... \ [A\vec{v}_n]_B\right]$

$= [P^{-1}(A\vec{v}_1) \ ... \ P^{-1}(A\vec{v}_n)]$

$= [(P^{-1}A)\vec{v}_1 \ ... \ (P^{-1}A)\vec{v}_n]$

$= P^{-1}A[\vec{v}_1 \ ... \ \vec{v}_n]$

$= P^{-1}AP$ or $[L]_B = {}_S P_B^{-1} [L]_S P_B$

> **Info — Similar Matrices**
>
> If $A$ and $B$ be $n \times n$ matrices s.t. $P^{-1}AP = B$ for some invertible matrix $P$, t
>
> 1. $A$ is said to be **similar** to $B$
> 2. rank $A$ = rank $B$
> 3. det $A$ = det $B$
> 4. tr $A$ = tr $B$ where tr $A$ is defined by tr $A = \sum_{i=1}^{n}(a_{ii})$ and is called the **trace** of a matrix

Note:

If $A$ is similar to $B$, $\exists P$ invertible s.t.

$$B = P^{-1}AP$$
$$PB = PP^{-1}AP$$
$$PB = AP$$
$$A = PBP^{-1}$$

## Eigenvalues and Eigenvectors

With $A = [L]$ and $P = [\vec{v}_1 ... \vec{v}_n]$

$D = \text{diag}(\lambda_1, \lambda_2, ..., \lambda_n) = \left[\lambda_1\vec{v}_1 \ \lambda_2\vec{2} \ ... \ \lambda_n\vec{v}_n\right]$

We want $D = P^{-1}AP \implies PD = AP$

Setting $AP = PD$, we get

$$A[\vec{v}_1 \ ... \ \vec{v}_n] = P[\lambda_1\vec{e}_1 \ ... \ \lambda_2\vec{e}_n]$$
$$[A\vec{v}_1 \ ... \ A\vec{v}_n] = [P(\lambda_1\vec{e}_1) \ ... \ P(\lambda_2\vec{e}_n)]$$
$$= [\lambda_1(P\vec{e}_1) \ ... \ \lambda_n(P\vec{e}_n)]$$
$$= [\lambda_1\vec{v}_1 \ ... \ \lambda_n\vec{v}_n] \qquad \text{(By Column Extraction Theorem)}$$

$\implies A\vec{v}_1 = \lambda_i\vec{v}_i, \forall i = 1, 2, ..., n$

Meaning that the effect of multiplying $A$ by $\vec{v}_i$ is the same as scaling $\vec{v}_i$ by some $\lambda_i$

> **Info — Eigenvalues, Eigenvectors, Eigenpairs**
>
> 1. Let $A$ be $n \times n$ matrix. If there exists a vector $\vec{v} \neq \vec{0}$ s.t. $A\vec{v} = \lambda\vec{v}$, then the scalar $\lambda$ is the **eigenvalue** of $A$ and $\vec{v}$ is called **eigenvectors** of $A$ corresponding to $\lambda$. The pair $(\lambda, \vec{v})$ is called an **eigenpair**
> 2. Let $L : \mathbb{R}^n \to \mathbb{R}^n$ be a linear operator. If there exists a vector $\vec{v} \neq \vec{0}$ s.t. $L(\vec{v}) = \lambda\vec{v}$, then $\lambda$ is caled an **eigenvalue** of $L$ and $\vec{v}$ is called an **eigenvector** of $L$ corresponding $\lambda$
> 3. A scalar $\lambda$ is an eigenvalue of a $n \times n$ matrix $A$ if and only if $C_{A(\lambda)} = 0$

How to find Eigenvalues

We need $\vec{v} \neq \vec{0}$ s.t. $A\vec{v} = \lambda\vec{v}$

$A\vec{v} - \lambda\vec{v} = \vec{0} \implies = (A - \lambda I)\vec{v} = \vec{0}$

To get non-zero $\vec{v}$ solution, $\det(A - \lambda I) = 0$

Examples:

1. Let $A = \begin{bmatrix}0 & 1\\ 2 & 1\end{bmatrix}, \vec{v} = \begin{bmatrix}1\\ 2\end{bmatrix}$

   $A\vec{v} = \begin{bmatrix}0 & 1\\ 2 & 1\end{bmatrix}\begin{bmatrix}1\\ 2\end{bmatrix} = \begin{bmatrix}2\\ 4\end{bmatrix} = 2\begin{bmatrix}1\\ 2\end{bmatrix}$ = 2 vec(v).

   $\lambda = 2$ is an eigenvalue of $A$ correspoind to the eigenvector $\vec{v} = \begin{bmatrix}1\\ 2\end{bmatrix}$

2. Let $A = \begin{bmatrix} 0 & 1 \\ 3 & -2 \end{bmatrix}$, find eigenvalues and eigenvectors of $A$

$$0 = \det\,(A - \lambda I) = \det\left(\begin{bmatrix} 0 & 1 \\ 3 & -2 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix}\right) = \begin{bmatrix} -\lambda & 1 \\ 3 & -2-\lambda \end{bmatrix}$$

$$= -\lambda(-2-\lambda) - 3 = (\lambda+3)(\lambda-1) = 0$$

$\implies \lambda_1 = 1, \lambda_2 = -3$

$A - \lambda_1 I = \begin{bmatrix} -1 & 1 \\ 3 & -3 \end{bmatrix} \sim \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix} \implies \vec{v}_1 = t\begin{bmatrix} 1 \\ 1 \end{bmatrix}, t \in \mathbb{R} \setminus \{0\}$

$A - \lambda_2 I = \begin{bmatrix} 3 & 1 \\ 3 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & \frac{1}{3} \\ 0 & 0 \end{bmatrix} \implies \vec{v}_2 = s\begin{bmatrix} -1 \\ 3 \end{bmatrix}, s \in \mathbb{R} \setminus \{0\}$

> **Info — Eigenspace**
>
> Let $A$ be a $n \times n$ matrix with eigenvalue $\lambda$. We call the nullspace of $A - \lambda I$ the **eigenspace** of $\lambda$, denoted $E_\lambda$
>
> Let $A$ be a $n \times n$ matrix with eigenvalue $\lambda_1$.
>
> - **Algebraic Multiplicity**
>  - The algebraic multiplicity of $\lambda_1$, denoted $a_{\lambda_1}$, is the number of times that $\lambda_1$ is a root of the characteristic polynomial $C(\lambda)$. That is, if $C(\lambda) = (\lambda - \lambda - 1)^k C_1(\lambda)$, where $C_1(\lambda_1) \neq 0$, then$a_{\lambda_1} = k$
> - **Geometric Multiplicity**
>  - The geometric multiplicity of $\lambda_1$, denoted $g_{\lambda_1}$, is the dimension of its eigenspace, $g_{\lambda_1} = \dim\left(E_{\lambda_1}\right)$
>
> 1. Let $A$ and $B$ be similar matrices, then $A$ and $B$ have the same characteristic polynomial, and hence the same eigenvalue.
> 2. If $A$ is a $n \times n$ matrix with the eigenvalue $\lambda_1$, then $1 \leq g_{\lambda_1} \leq a_{\lambda_1}$

Examples:

Let $A = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$. Find the eigenvalues of $A$, bases for the corresponding eigenspace, state the geometric and algebraic multiplicity of each eigenvalue.

$$0 = \det(A - \lambda I) = \det\left(\begin{bmatrix} -\lambda & 1 & 1 \\ 1 & -\lambda & 1 \\ 1 & 1 & -\lambda \end{bmatrix}\right)$$

$$= -\lambda\,\det\begin{bmatrix} -\lambda & 1 \\ 1 & -\lambda \end{bmatrix} - \det\begin{bmatrix} 1 & 1 \\ 1 & -\lambda \end{bmatrix} + \det\begin{bmatrix} 1 & -\lambda \\ 1 & 1 \end{bmatrix} = -\lambda(\lambda-1)(\lambda+1) + (\lambda+1) + (\lambda+1)$$

$$= -(\lambda+1)(\lambda^2 - \lambda - 2) = -(\lambda+1)^2(\lambda-2) \implies$$

$\lambda_1 = -1, \lambda_2 = 2$

$$A - \lambda_1 I = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \implies s\begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} + t\begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \implies B_{\lambda_1}\left\{\begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}\right\}.$$

Bases for $\lambda_1 = \text{Span } B_{\lambda_1}$

$$A - \lambda_2 I \implies r\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \implies B_{\lambda_2}\left\{\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}\right\}.$$

Bases for $\lambda_2 = \text{Span } B_{\lambda_2}$

For $\lambda_1 = -1, a_{\lambda_1} = 2, g_{\lambda_1} = 2$

For $\lambda_2 = 2, a_{\lambda_1} = 1, g_{\lambda_1} = 1$

**Proof**

Eigenspace Theorem 1

Suppose that $\exists P(\text{invertible})$ s.t. $B = P^{-1}AP$, then

$$B - \lambda I = P^{-1}AP - \lambda I = P^{-1}AP - \lambda P^{-1}P$$
$$= P^{-1}AP - \lambda P^{-1}IP = P^{-1}(AP - \lambda IP) = P^{-1}((A - \lambda I)P) = A - \lambda I$$

Thus $\det\ B - \lambda I = \det\ A - \lambda I$

□

## Diagonalization

**Theorem — Diagonalization Theorem**

A $n \times n$ matrix $A \in M_{n\times n}(\mathbb{R})$ is said to be **diagonalizable** if $A$ is similar to a diagonal matrix $D \in M_{n\times n}(\mathbb{R})$. If $P^{-1}AP = D$, then $P$ diagonalizes $A$.

- A $n \times n$ matrix $A$ is diagonalizable (over $\mathbb{R}$) if and only if there exists a basis $\{\vec{v}_1, ..., \vec{v}_n\}$ for $\mathbb{R}^n$ of eigenvector of $A$
- If $A$ is a $n \times n$ matrix with eigenpairs $(\lambda_1, \vec{v}_1), ..., (\lambda_k, \vec{v}_k)$ where $\lambda_i \neq \lambda_j$ for $i \neq j$, then $\{\vec{v}_1, ..., \vec{v}_k\}$ is linearly independent.
- If $A$ is a $n \times n$ matrix with distinct eigenvalues $\lambda_1, ..., \lambda_k$ and $B_i = \left\{\vec{v}_{i,1}, ..., \vec{v}_{i,g_{\lambda_i}}\right\}$ is a basis for the eigenspace of $\lambda_i$ for $1 \leq i \leq k$, then $B_1 \cup B_2 \cup ... \cup B_k$ is a linearly independent set
- If $A$ is a $n \times n$ matrix with $n$ distinct eigenvalues, then $A$ is diagonalizable.

**Diagonalizability Test**

If $A$ is a $n \times n$ matrix whose characteristic polynomial factors as

$$C_A(\lambda) = (\lambda - \lambda_1)^{a_{\lambda_1}}...(\lambda - \lambda_k)^{a_{\lambda_k}}$$

where $\lambda_1, ..., \lambda_k$ are the distinct eigenvalues of $A$, then $A$ is diagonalizable if and only if $g_{\lambda_i} = a_{\lambda_i}$ for $1 \leq i \leq k$.

**Lemma**

If $A$ is a $n \times n$ matrix with $n$ distinct eigenvalues, then $A$ is diagonalizable.

**Theorem**

- If $\lambda_1, ..., \lambda_n$ are all the $n$ eigenvalues of a $n \times n$ matrix $A$(repeated according to algebraic multiplicity), then

$$\det\ A = (\lambda_1)...(\lambda_n) \text{ and } \operatorname{tr}\ A = \lambda_1 + ... + \lambda_n$$

Example:

Let $A = \begin{bmatrix} 2 & 2 & 2 \\ 2 & -2 & 0 \\ 2 & 0 & 1 \end{bmatrix}$

Diagonalize $A$ if possible, that is, for an invertible matrix $P$ and a diagonal matrix $D$ s.t.

$PAP^{-1} = D$

$C(\lambda) = 0 \implies 0 = \det(A - \lambda I) = \det \begin{bmatrix} 2-\lambda & 2 & 2 \\ 2 & -2-\lambda & 0 \\ 2 & 0 & 1-\lambda \end{bmatrix} = -\lambda(\lambda - 4)(\lambda + 3)$

$\implies \lambda_1 = 0, \lambda_2 = 4, \lambda_3 = -3$ are eignvalues of $A$

Since $A$ has 3 distinct eigenvalues, thus it is diagonalizable

$\lambda_1 = 0 \implies \begin{bmatrix} 2 & 2 & 2 \\ 2 & -2 & 0 \\ 2 & 0 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & \frac{1}{2} \\ 0 & 1 & \frac{1}{2} \\ 0 & 0 & 0 \end{bmatrix} \implies \vec{v} = s \begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix}, s \in \mathbb{R}$

$B_1 = \left\{ \begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix} \right\}$ is a basis for $E_{\lambda_1}$

$\lambda_2 = 4 \implies \begin{bmatrix} -2 & 2 & 2 \\ 2 & -6 & 0 \\ 2 & 0 & -3 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -\frac{3}{2} \\ 0 & 1 & -\frac{1}{2} \\ 0 & 0 & 0 \end{bmatrix} \implies \vec{v} = t \begin{bmatrix} 3 \\ 1 \\ 2 \end{bmatrix}, t \in \mathbb{R}$

$B_2 = \left\{ \begin{bmatrix} 3 \\ 1 \\ 2 \end{bmatrix} \right\}$ is a basis for $E_{\lambda_2}$

$\lambda_3 = -3 \implies \begin{bmatrix} 5 & 2 & 2 \\ 2 & 1 & 0 \\ 2 & 0 & 4 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & -4 \\ 0 & 0 & 0 \end{bmatrix} \implies \vec{v} = q \begin{bmatrix} -2 \\ 4 \\ 1 \end{bmatrix}, q \in \mathbb{R}$

$B_3 = \left\{ \begin{bmatrix} -2 \\ 4 \\ 1 \end{bmatrix} \right\}$ is a basis for $E_{\lambda_3}$

Let $P = [\vec{v}_1 \ \vec{v}_2 \ \vec{v}_3] = \begin{bmatrix} 1 & 3 & -2 \\ 1 & 1 & 4 \\ -2 & 2 & 1 \end{bmatrix}, D = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & -3 \end{bmatrix}$

## Powers of Matrices

> **Info — Powers of Matrices**
> Let $A$ be a $n \times n$ matrix. If there exists a matri $P$ and diagonal matrix $D$ s.t. $P^{-1}AP = D$, then
>
> $$A^k = PD^kP^{-1}$$

**Proof**

$D = P^{-1}AP \implies A = PDP^{-1} \implies$

$A^k = (PDP^{-1})^k = (PDP^{-1})(PDP^{-1})(PDP^{-1})...(PDP^{-1}) = PD(P^{-1}P)D(P^{-1}P)...(P^{-1}P)DP^{-1}$

$= PD^kP^{-1}$

□

Example:

Let $A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$. Diagonalize $A$ and compute $A^{307}$

It can be show that $\lambda_1 = -1, \lambda_2 = 3$ are eigenvalues of $A$ with corresponding eigenvectors $\vec{v}_1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \vec{v}_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ respectively.

Let $P = [\vec{v}_1 \ \vec{v}_2] = \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}, D = \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & 3 \end{bmatrix}$

Then $D = P^{-1}AP, P^{-1} = \frac{1}{2}\begin{bmatrix} -1 & -1 \\ 1 & 1 \end{bmatrix}, D^{307} = \begin{bmatrix} -1 & 0 \\ 0 & 3^{307} \end{bmatrix}$

$A^{307} = \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}\begin{bmatrix} -1 & 0 \\ 0 & 3^{307} \end{bmatrix}\frac{1}{2}\begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} = \frac{1}{2}\begin{bmatrix} -1+3^{307} & 1+3^{306} \\ 1+3^{307} & -1+3^{307} \end{bmatrix}$

## Complex Numbers

Definitions for Complex Numbers

$$\mathbb{C}^n = \left\{ \begin{bmatrix} z_1 \\ z_2 \\ ... \\ z_n \end{bmatrix} \mid z_1, z_2, ..., z_n \in \mathbb{C} \right\}$$

Additional and scalar multiplication in the same ways as $\mathbb{R}^n$

$M_{m\times n}(\mathbb{C})$ is the set of all $m \times n$ matrices with complex entries.

For $A \in M_{m\times n}(\mathbb{C})$, $\overline{A}$ is defined by $\left(\overline{A}_{ij}\right) = \overline{a}_{i_j}$

For $\vec{z} \in \mathbb{C}^n$, $\overline{\vec{z}} = \begin{bmatrix} \overline{z_1} \\ ... \\ \overline{z_n} \end{bmatrix}$

$A = [\vec{a}_1\ \vec{a}_2\ ...\ \vec{a}_n], \overline{A\vec{z}} = \overline{z_1\vec{a}_1 + ... + z_n\vec{a}_n} = \overline{z_1\vec{a}_1} + ... + \overline{z_n}\vec{a}_n = \overline{z_1}\overline{\vec{a}_1} + ... + \overline{z_n}\overline{\vec{a}_n}$

$$= \begin{bmatrix} \overline{\vec{a}_1} & ... & \overline{\vec{a}_n} \end{bmatrix} \begin{bmatrix} \overline{z_1} \\ ... \\ \overline{z_n} \end{bmatrix}$$

Examples:

1. Let $A = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$, diagonalize $A$ over $\mathbb{C}$

   0 $\det(A - \lambda I) = \det \begin{bmatrix} 1-\lambda & -1 \\ 1 & 1-\lambda \end{bmatrix} = (1-\lambda)^2 + 1$

   $-1 = (1-\lambda)^2 \Longrightarrow 1 - \lambda = \pm i \Longrightarrow \lambda = 1 \pm i$

   $A - \lambda_1 I = \begin{bmatrix} i & -1 \\ 1 & i \end{bmatrix} (-i)R_1 \sim \begin{bmatrix} 1 & i \\ 1 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \Longrightarrow x_1 = -ix_2 = -it \Longrightarrow \vec{x} = t\begin{bmatrix} -i \\ 1 \end{bmatrix}, t \in \mathbb{C}$

   $B_1 = \left\{ \begin{bmatrix} -i \\ 1 \end{bmatrix} \right\}$ is a basis for $E_{\lambda_1}$

   Similarily, $A - \lambda_2 I$ have the same process.

   $B_2 = \left\{ \begin{bmatrix} i \\ 1 \end{bmatrix} \right\}$ is a basis for $E_{\lambda_2}$

   Let $P = \begin{bmatrix} -i & i \\ -1 & 1 \end{bmatrix}, D = \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix} = \begin{bmatrix} 1-i & 0 \\ 0 & 1+i \end{bmatrix}$

   These satisfy $D = P^{-1}AP$

2. Compute $A^{100}$ from (1)

   $A^{100} = PD^{100}P^{-1}$

   $P^{-1} = \frac{1}{-2i}\begin{bmatrix} 1 & -i \\ -1 & -i \end{bmatrix}$

   $\overset{\frac{1}{i}=-i}{=} \frac{1}{2}\begin{bmatrix} i & 1 \\ -i & 1 \end{bmatrix}$

   $D^{100} = \begin{bmatrix} \sqrt{2}(\cos 25\pi - i \sin 25\pi) & 0 \\ 0 & \sqrt{2}(\cos 25\pi + \sin 25\pi) \end{bmatrix} = \begin{bmatrix} -2^{50} & 0 \\ 0 & -2^{50} \end{bmatrix}$

   $A^{100} = \begin{bmatrix} -i & i \\ 1 & 1 \end{bmatrix} \begin{bmatrix} -2^{50} & 0 \\ 0 & -2^{50} \end{bmatrix} \frac{1}{2} \begin{bmatrix} i & 1 \\ -i & 1 \end{bmatrix} = \begin{bmatrix} -2^{50} & 0 \\ 0 & -2^{50} \end{bmatrix}$

📐 **Theorem — Theorem**

Let $A \in M_{n\times n}(\mathbb{R})$. If $\lambda$ is a non-real eigenvalue of $A$ with corresponding eigenvector $\vec{z}$, then $\overline{\lambda}$ is also an eigenvalue of $A$ with eigenvector $\overrightarrow{\overline{z}}$

Also, if $n$ is odd, $A$ has at least one real eigenvalue.

**END OF FINAL**